

Assignment - 3

Nice try, no more Claude reports indeed! :D /s

Also, using pen and paper after a long time, back to my roots - thanks!

(Nvm, I'm just joking lol)

Q1 Why does the notebook download imagery live from a server instead of shipping a satellite image file with the notebook? Give one practical reason.

→ We are downloading images live from a server instead of shipping a satellite image file with the notebook because this way we avoid including large files in the notebook and download only the required tiles for the selected area.

One practical reason also is that with the help of the server links, we can choose any location by setting the coordinates of our choice instead of being limited to one preloaded image.

Q2 If you increased the zoom from 18 to 19, what happens to (a) the number of tiles downloaded and (b) how clearly the model can see small rooftops?

→ (a) Since the geographic area the image has to cover remains same, zooming will thus increase the image resolution and hence, more tiles get downloaded.

(b) Because of the zoomy, small rooftops appear clearer and larger, so the model can now see them better to detect and distinguish between other similar rooftops.

Q3 We filter footprints with confidence ≥ 0.65 . What happens to our building count if we lowered this to 0.4? What if we raised it to 0.9? Which errors does each choice risk (missing real buildings vs counting fake ones)?

Ans If we lower this to 0.4, then we risk false positives, that is even those things that aren't an actual building will be classified as a building. More detections will be accepted, reducing the chance of missing real buildings but several fake ones will be counted as real.

If we raise it to 0.9, the threshold value becomes so high that it will miss several real buildings, those which are real but have a less confidence score than which is undesirable. Thus, such a high threshold value will induce the error of missing actual real buildings.

Q5 Fine-tuning continues training from the pretrained weights instead of starting from scratch. Why is that faster and needs less data than training a brand-new model?

→ Fine-tuning is faster because the model has already learned many general features of a building, such as edges, shapes, and rooftops, from pretrained weights. Instead of learning these basic patterns again from scratch, it only needs to adapt them to the appearance of Indian buildings. This requires less data because the model is refining the

Knowledge it already knows rather than learning everything from the beginning.

Q6 Even after fine-tuning, best india-pt only scores around 0.27 mAP on Indian data - far below the 0.708 it got in the sandbox. But the detection images look reasonable. How can the number be low while the picture looks good? (Hint: think about what mAP measures, and how good the "answer key" labels are.)

→ Because the model still misses many buildings because of small rooftops, irregular shape and closely packed buildings. However, the detection images still look reasonably good because the model successfully detects several distinct buildings. The mAP remains low because it is based on the area under the Precision-Recall curve, both the parameters are affected by every missed building and inaccurate bounding box, and also any imperfections in the answer key labels can further reduce the score. Thus, the visual output can appear useful while the overall evaluation metric is much lower.

Q7 We use the count of buildings per cluster as a "density score." For our next goal (finding where most people live :::: more to this later:)), what does a building count miss? (Hint: is every building the same? Think about a 1-storey house vs a 20-floor tower.)

→ A building count only tells us how many buildings are present in a particular area. It does not account for differences between buildings, such as a 1-storey house and a 20-floor apartment tower, even though they can accommodate vastly different numbers of people. Since

Satellite images mainly show small rooftop and top view, the building count estimates only building density, not population density. Therefore, building count alone is not sufficient to determine where most people live which is an important factor for our analysis to find the hot zone in the city to set up our dark stores.

Q8 (most important): You spent Session 1 training a model and Session 2 fine-tuning it - then we use Open Buildings for the real answer anyway. Was model building a waste of time? Argue your view in 4-5 sentences. (There's no single right answer - we want your reasoning about when to build your own tool vs use an existing one.)

→ Model-building wasn't a waste of time. Instead, it gave valuable insights to how object detection models work, and deepened my understanding of YOLO, transfer learning, evaluation metrics etc. Fine tuning with best-india.pt was quite good, it was good to see our model's performance in the actual task we wanted to do and what it looked like. Building our own tool is a good learning experience that gives us knowledge of how real-life applications are being carried out, but using already existing ones (such as Open Buildings) is more efficient, accurate and time-saving for real-life purposes because they have been already tried and tested and are more reliable for our analysis.

BLACK PEN INK RAN OUT, SORRY!

CODE PART:

→ NO, I
CAN'T :(

Question 1 (write 2-3 sentences): In your own words, why did the sandbox model fail on lucknow even though it scored 0.7-0.8 mAP in session 1?

→ Sandbox model worked on US and European cities where it learned patterns that buildings are large, regular, well-separated, with clean roof edges which is the norm there. But the situation is different in India where the buildings are small, irregularly packed, wall-to-wall, and built of different designs and varied materials. Because of this domain gap, it fails to detect similar patterns it had already learnt in lucknow and thus, we had to fine tune it.

Question 2: Roughly, how many buildings did your model detect in your patch? where did it miss the most?

→ My model detected 10786 buildings in my patch. On inspecting the densest tile, I could notice that the model missed out on the densest slums where the rooftops are quite small and so wall to wall packed that it struggled to identify each one of them accurately.

Question 3: Open Buildings almost certainly has more buildings than your model. Why? And given that - when would a real data scientist choose to use an existing dataset instead of running their own model?

→ Open Buildings found 908,029 buildings, far more than our model which found 10,786 buildings because it was created using a much larger and more powerful model trained on vast amounts of high-resolution satellite imagery throughout India. As a result, it is much better adapted to Indian building patterns and detects buildings more accurately than our fine-tuned model.

A real data scientist would choose an existing data set instead of running their own model when it is reliable, well-tested, and already solves the problem accurately, as it saves time and computational resources. However building our own custom model is useful when no suitable dataset exists or when the task requires a specialised solution or for learning purposes.

END !!

MY HANDS
HURT 😞



- Got brownie points for overacting.

But in all seriousness, I was
BORED!!

Who gives a handwritten
assignment seriously?

} FEEDBACK

- Yoshinori Kashyap
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